

Osama Ali Shah (Certified AI Engineer)

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Profile:

I am an **AI Engineer** with expertise in **Machine learning, Deep learning, Computer Vision (Image & Video Analysis) and Generative AI (LLMs)**. Skilled in **data analysis, visualization, and model development** using tools like **NumPy, Pandas, OpenCV, TensorFlow, and PyTorch**. Experienced in building **LLM agents with LangChain, LangGraph, MCP** and deploying AI solutions with **Flask, FastAPI, and Streamlit**. Passionate about creating intelligent, scalable systems that solve real-world problems.

- Skills & Interest:**
- **Data Analysis & Manipulation:** NumPy, Pandas
 - **Data Visualization:** Matplotlib, Seaborn, Plotly
 - **Generative AI / LLMs:** LangChain, LangGraph, MCP
 - **Machine Learning:** scikit-learn, XGBoost, LightGBM
 - **Deep Learning:** TensorFlow, Keras, PyTorch
 - **Computer Vision:** OpenCV, CNNs, YOLO, Image Classification, Object Detection, Video Analysis
 - **Web Development:** Flask, FastAPI, Streamlit
 - **Web Scraping:** BeautifulSoup, Selenium, Requests, Apify
 - **Automation & Workflow Orchestration:** n8n
 - **Database & ORM:** SQLite, Postgres

- Professional Experience:**
- AI Engineer – SAT Impact Solutions*
- OCT 2025 – Ongoing*
- Develop and deploy AI and Machine Learning models for real-world applications.
 - Work on Natural Language Processing (NLP) and Deep Learning projects.
 - Perform data collection, preprocessing, and model evaluation.
 - Integrate AI models into web-based systems and enterprise solutions.
 - Collaborate with development teams to ensure efficient model deployment.
 - Research and implement state-of-the-art AI techniques to improve system performance.

- AI Engineer – AI Data House*
- 3 Months*
- Developed and deployed **LLM-powered applications** for conversational AI, document analysis, and intelligent automation.
 - Built **multi-agent systems** using LangChain and LangGraph to manage complex workflows and task orchestration.
 - Designed and integrated **chatbots and custom AI agents** with advanced features such as PDF Q&A, resume analysis, and advisory conversations.
 - Implemented **end-to-end AI pipelines**, from data processing and model integration to real-world deployment.
 - Containerized and deployed AI solutions with **Docker**, ensuring scalability, portability, and efficient deployment workflows.

- Professional Projects:**
- **Podcast AI Agent Development**

- Designed and deployed an LLM-based RAG system to process podcast transcripts, enabling content analysis, automated question generation, and advisory-style interactions aligned with the podcast's brand voice.
 - Implemented the solution using LangChain, FAISS, and LLM fine-tuning to deliver a scalable, knowledge-driven engagement platform.
- **Multi-Agent Chatbot with LangGraph**
 - Developed a LangGraph-based multi-agent chatbot integrating multiple workflows, enabling natural conversations with specialized agents for career counseling, resume evaluation, and document querying.
 - Implemented PDF upload and real-time Q&A functionality to demonstrate practical, user-focused applications of LLMs and multi-agent systems.
- **AI-Powered Smart Vending Machine**
 - Developed an AI-powered vending machine using RAG to recommend protein supplements and vitamin products based on user context and preferences.
 - Integrated multimodal inputs (face recognition, text, voice) and a structured nutrition database to deliver personalized and accurate health recommendations.
- **Human Activity Recognition**
 - Developed a CNN-LSTM-based computer vision system to recognize human activities from video data using spatial-temporal features.
 - Implemented OpenCV and YOLO for real-time human detection and video-based activity analysis
- **AI Engine for Dental Disease Diagnosis**
 - Developed an AI-based diagnostic engine to detect dental diseases using a Kaggle medical image dataset, involving data preprocessing and preparation for model training.
 - Designed and implemented a Convolutional Neural Network (CNN) to classify dental conditions, enabling faster and more accurate diagnosis and demonstrating the potential of deep learning in healthcare.

Education:

Abdul Wali Khan University Mardan.

Sept 2021 – July 2025

- Degree: Bachelor of Science in Computer Science
- Current CGPA: 3.48/4.0

Certifications:

Coursera (2022-2024):

- Python for Data Science, AI & Development
- Machine Learning Specialization (3 courses)
- Deep Learning Specialization (5 courses)
- Natural Language Processing with Classification and Vector Spaces (4 courses)

**GreatLearning
(2022-2023):**

- NumPy
- Pandas
- Matplotlib
- Seaborn
- SpaCy

Udemy (2024-2025):

- Complete Generative AI Course with LangChain & Hugging Face
- LangGraph
- Post Processing of LLMs
- Introduction to reinforcement Learning (HuggingFace)